Agenda – 120 mins

- Zoom Ground Rules
- Intro
- Challenge
- Workshop
- Slido Q & A
- Open Mic

Ground Rules

Slido:

JETBRAINS

DECODING
DATA SCIENCE

AI Accelerator Bootcamp

Workshop 1: Project Scope, Data, OS and LLM

From AI Curious to AI Builder

Build a real enterprise HR chatbot using retrieval, grounded generation and a production-style stack.

Mohammad Arshad • Decoding Data Science

Day 1

DECODING DATA SCIENCE

Clarity before coding

Scope | Data | Workflow | LLM

WELCOME

Welcome to the Bootcamp

A 3-day practical journey from content consumer to AI builder.

- Understand how a real AI application is structured end to end
- Work through a practical enterprise use case, not a toy demo
- Connect problem, data, retrieval, model and interface into one system
- Leave with a stronger mental model for building your own AI apps

What you achieve in 3 days

Day 1 Scope the project, understand the data, workflow and the LLM setup.

Day 2 Build retrieval thinking: chunking, embeddings, indexing and vector search.

Day 3 Connect the full RAG flow, interface the app and get it demo ready.

AGENDA

Today's Agenda

1. Why many people stay stuck at the AI-curious stage
2. What a real AI application actually needs
3. The project we are building: DDS HR Enterprise Chatbot
4. Why RAG matters in enterprise AI
5. Architecture, stack and LLM setup
6. Day 1 checkpoint and what comes next

Today's theme

Clarity before coding

The goal today is simple: make the system feel understandable, practical and buildable.

Start with scope • data • workflow • LLM

MINDSET

Why Most People Stay Stuck at “AI Curious”

The common loop

- Consume tutorials and tool demos
- Try prompts without a clear use case
- Jump into code before defining the problem
- Ignore the role of data and retrieval
- End up with a vague demo and more confusion

The gap is not motivation.
It is structure.

Builders win because they work from a simple sequence:
problem → data → retrieval → generation → interface.

AI Curious

Asks which tool is best.

AI Builder

Asks what system is needed.

From Tool User to System Thinker

A real AI app is more than a model. It is a workflow built around a user problem and the right data.

FOUNDATION

What a Real AI App Actually Needs

Use case + data + retrieval + LLM + interface + evaluation = a useful AI application

Problem

A clear user and a clear task

Data

Relevant source documents and knowledge

Retrieval

A way to find the right context at the right moment

LLM

Reasoning and grounded response generation

Interface

A simple way for users to interact

Evidence

A way to judge quality and improve

PROJECT

The Project: DDS HR Enterprise Chatbot

- Grounded on company documents rather than generic model memory
- Useful for repetitive internal queries like leave, benefits and onboarding
- Built as an end-to-end RAG workflow, not just a prompt demo
- Good portfolio example because it reflects a real enterprise pattern

We are building a practical enterprise-style assistant that answers HR questions from internal documents and policy content.

Typical questions

What is the annual leave policy?

How does remote work approval work?

What benefits are available after probation?

Where can new hires find onboarding guidance?

DECODING DATA SCIENCE

What Is RAG?

Retrieval-Augmented Generation

1. Retrieve

Find the most relevant chunks from your documents.

2.Augmentation

Pass that context into the model prompt.

3. Generate

Produce an answer based on retrieved evidence.

Instead of answering only from model memory, the system first retrieves relevant company knowledge and then generates a grounded response.

Without RAG

- Only general model knowledge
- Higher hallucination risk
- Weak grounding in company data

With RAG

- Answers based on internal documents
- More relevant and useful responses
- Much better fit for business workflows

ARCHITECTURE

How the System Works

Think of this as a workflow, not just a model.

Docs
Policies and source files

→

Prepare
Chunk and clean content

→

Embed
Create semantic vectors

→

Store
Save vectors in Pinecone

→

Ask
User sends a question

→

Retrieve
Find the best matching chunks

→

Answer
LLM writes a grounded response

The system becomes useful because it retrieves the right company knowledge before the model writes the answer.

STACK

Stack Overview and LLM Setup

LlamaIndex

Retrieval workflow and orchestration

Pinecone

Vector database and semantic search

Embeddings

Represent document chunks and user queries

OpenAI LLM

Reasoning and grounded answer generation

Gradio

Simple interface for the final experience

For this bootcamp we use OpenAI embeddings plus an OpenAI LLM so participants can focus on architecture, retrieval logic and practical application design.

WORKSHOP 1

DECODING DATA SCIENCE

Today’s Focus: Scope, Data, OS and LLM

Project scope — Who is the user and what exact problem are we solving?

Data thinking — Which documents do we need and what quality do they have?

OS / system flow — How do retrieval, generation and interface fit together?

LLM choice — What model and embeddings setup are we using for the workflow?

THINKING

Questions We Must Answer Before Building

Scope

- Who is the primary user?
- What exact problem are we solving?
- What should the assistant answer?
- What should it not answer?
- What makes this useful?

Data

- Which documents matter for version 1?
- Are they clean and relevant?
- Do they need chunking or preprocessing?
- What gaps may weaken retrieval?

OS + LLM

- How does the workflow move from query to answer?
- Where does retrieval happen?
- Which model and embeddings setup will we use?
- How will we judge success?

CHECKPOINT

Day 1 Checkpoint

By the end of today, every participant should be able to define the project clearly.

Checkpoint template

- Project title
- Primary user
- Problem statement
- Data source(s)
- Expected output
- Why it matters
- Definition of success
- LLM / embeddings setup

HR Policy Assistant Example

User: Employees and HR team

Problem: Fast, accurate answers from policy documents

Data: Handbook, leave policy, benefits PDF

Output: Grounded answer with the right context

Success: Helpful responses to common HR questions

NEXT

What’s Coming in Day 2

Tomorrow we turn the architecture into a retrieval pipeline.

Chunking

Break documents into meaningful passages

Embeddings

Represent chunks and queries semantically

Indexing

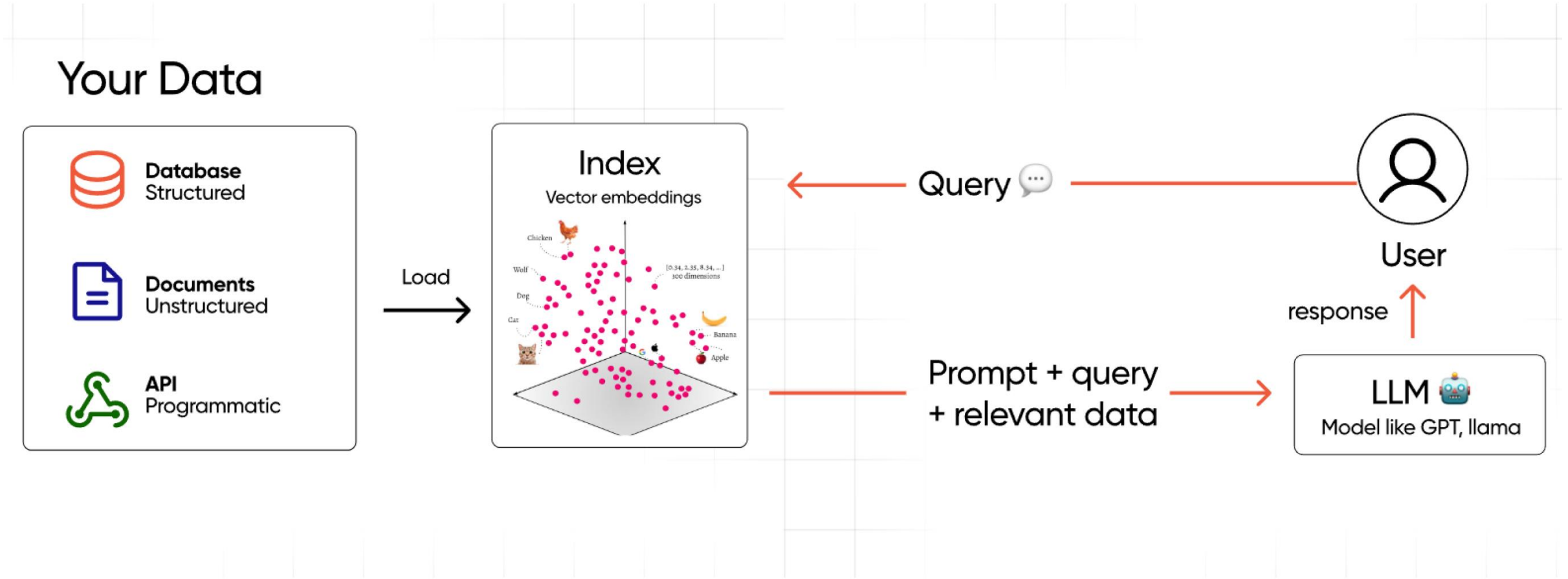
Organize data for fast retrieval

Pinecone

Store and search vectors

LlamaIndex

Wire the retrieval workflow together


Your Data
Database
Structured
Documents
Unstructured
API
Programmatic
Load
Index
Vector embeddings
Chicken
Wolf
Dog
Cat
Banana
Apple
Query
User
response
Prompt + query
+ relevant data
LLM
Model like GPT, llama

NEXT

What’s Coming in Day 3

Then we connect everything into a usable end-to-end assistant.

1. Connect retrieval to the LLM prompt
2. Generate grounded responses
3. Build the Gradio interface
4. Test, refine and demo the chatbot

Outcome

A practical portfolio-ready RAG assistant

CLOSE

Key Takeaway and Next Step

A vague AI idea creates a vague AI app.

Start with clarity on the problem, the data, the workflow and the model setup — then building becomes much easier.

- Complete the Day 1 checkpoint
- Review your project idea and documents
- Come ready for retrieval workflow on Day 2
- Bring questions and edge cases to the next session

DECODING DATA SCIENCE

Q&A

Let's move from clarity to prototype.